

Mohamed Zakaria Mohamed Ahmed

Software engineer working on ERP systems , AI agents and AI automation, with research experience in Italy and Rwanda, driven by a strong interest in AI research for African languages and accessibility solutions

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RESEARCH EXPERIENCE

AI Research & Engineering Intern (Erasmus+) (July 2026 – August 2026)

Università degli Studi di Parma — Parma, Italy

Project: Evaluating a Quantized ECG Heartbeat Classifier for Edge Devices

[Research write-up](#) | [Code](#)

- **Problem:** Cloud-based ECG analysis sends sensitive heart recordings off-site to a remote server, exposing patient data outside the clinic's control.
- **Solution:** An offline system that reads a heartbeat from a strip image and classifies it locally (~18 ms per image on one CPU core).
- **Research:** Tested on 1,000 ECG images from the ECG5000 dataset: **92.7% correct**, though rare heartbeat types were often missed due to limited examples.
- **Next:** Deploy on a microcontroller, a low-cost chip that could run inside a portable ECG device without a computer or internet.

Software Engineering Intern (March 2026 – May 2026)

Digital Umuganda — Kigali, Rwanda

Project: VisionTTS: Describing Scenes Aloud in Kinyarwanda for People with Visual Impairments

[Research write-up](#) | [Code](#) | [Demo](#)

- **Problem:** More than 158,000 Rwandans live with a visual impairment (2022 census), and leading assistive glasses (Envision, OrCam) cost \$599–\$4,000+ and do not support Kinyarwanda.
- **Solution:** A system that describes a scene or reads printed text aloud in Kinyarwanda from a camera image, built by combining three existing open AI models.
- **Research:** Chose compressed model versions to fit a small computer's memory. The working prototype took 31.5–43 s per answer in informal tests, which is far too slow for everyday use.
- **Next:** Move it onto low-cost smart glasses (3D parts designed, not yet built) so it works offline, make it faster, and measure its quality with standard tests.

Project: Kinyarwanda Sign Language (KSL) Gesture Recognition

[Research write-up](#) | [Demo](#)

- **Problem:** About 70,000 Rwandans are deaf or speech-impaired, and few people around them understand Kinyarwanda Sign Language (KSL). No public KSL dataset or tool existed.
- **Solution:** An offline system that recognises KSL signs from a camera and speaks them in Kinyarwanda, using a tiny 6.9 KB model small enough for a low-cost wearable.
- **Research:** Recorded 36,373 examples of 10 signs myself and trained the model on hand shape alone: **~98.5% correct**, likely optimistic since test and training frames came from the same recordings.
- **Next:** Recognise signs that use movement or both hands, and fit the model into a wearable smart glove.

WORK EXPERIENCE

Software Engineer (July 2026 – Present)

Sokrab Group — Kigali, Rwanda

- Implement ERP systems for client organisations, mapping their business processes into the system.
- Build AI automation workflows that reduce manual, repetitive tasks in day-to-day business operations. [Demo](#)
- Support coworkers in adopting AI tools in their daily work.

EDUCATION

BSc in Computer Science (Software Engineering)

INES-Ruhengeri, Rwanda — Graduation: October 2026

- **Cumulative GPA: 4.0/5.0 (Upper Second Class Honours)**
- **Earned a 5.0/5.0 GPA in every semester of my final two years.**

- *Quantitative coursework:* Probability and Statistics (MAT 1202), Operational Research (CSC 1206), Data Mining and Data Warehousing (SWE 3408), Artificial Intelligence (SWE 3513), Physics for Engineers (PHY 1201)

LEADERSHIP & COMMUNITY IMPACT

Team Leader, Code for Impact 2025 (2025)

Mastery Hub Rwanda — Kigali, Rwanda

- Placed **6th** and ranked among the **Top 10 startups** for making Kinyarwanda speech technology easier to use. We deployed Meta's open-source Kinyarwanda text-to-speech model as a fast service any application could call, and demonstrated it in a smart glove that turns sign language into speech.

Volunteer (August 2026 – Present)

Rural Social Industrial Complexes (RSIC) — Sudan

- Software Engineer Volunteer Lead a team of 3 volunteers bringing AI to a Sudanese non-profit initiative that supports rural communities through local industry and social development.
- Built the organisation's first **retrieval-augmented generation (RAG)** chatbot, which answers visitors' questions using its own documents.

Workshop Speaker (July 2026)

Sudanese Youth Club (SYC)

- Made Generative AI and AI agents approachable for high school students, and gave them practical first steps to start learning AI early.

Workshop Speaker (2026)

Nile Tech Community, Kigali Independent University (ULK) — Kigali, Rwanda

- Shared my own path from university student to working software engineer in Rwanda with first-year and graduating Computer Science students, and encouraged them to build projects and start their careers early.

Workshop Speaker (March & August 2026)

Sudanese Students Online Summit (SSOS)

- Led two sessions for Sudanese students: one mapping software engineering career paths (Backend, Data, Mobile, Cybersecurity) and one on securing internships, scholarships and international opportunities. Received a certificate of appreciation.

TECHNICAL SKILLS

- **Generative AI:** LLMs, Vision-Language Models, RAG, LangChain, Model Context Protocol (MCP), OpenAI Agents SDK, Google ADK, Gemini API
- **Programming:** Python (main language), JavaScript (Node.js), SQL
- **Backend & APIs:** RESTful API design, FastAPI, Flask, Express.js, WebSockets
- **Databases:** PostgreSQL, MongoDB, ChromaDB (vector database)
- **ERP & Automation:** Odoo, AI workflow automation
- **Tools:** Git, GitHub Actions (CI/CD), Docker, Postman